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The Organization of the Rat Motor Cortex: A Microstimulation Mapping Study

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1. INTRODUCTION

Although the location and overall somatotopic organization of the rat motor cortex have been known for some time[19,59], the detailed internal organization of its larger subdivisions (the forelimb area, for example) has not yet been described. Such a study seems warranted because of renewed interest in the rat motor cortex prompted by several findings. First, Elger et al.[16] have reported that electrical stimulation of the rat motor cortex can produce *monosynaptic* excitatory postsynaptic potentials (EPSPs) in cervical

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enlargement motorneurons; such monosynaptic connections had previously been found only in primates and racoons[6,29]. Second, both neuroanatomical[21,57] and electrical stimulation[42] studies have demonstrated what appear to be multiple representations of the forelimb and other body parts in rat frontal and parietal cortex. Finally, the cytoarchitectonic boundaries defined in rat sensorimotor cortex have been shown to correlate well with physiologically defined borders within and around this region[15,49]. The present study was undertaken to provide a complete, detailed stimulation map of the rat 'motor cortex' correlated with cortical cytoarchitectonic and functional borders.

2. MATERIALS AND METHODS

2.1. *Animal preparation*

Thirty-two adult Long–Evans hooded rats (250–500 g) were studied, each of which was anesthetized with ketamine HCl (100 mg/kg i.p., with supplemental doses of 1/4 the initial dose as needed). The hair covering the right forelimb and hindlimb was clipped to permit better observation of evoked movements, and then the animal was placed in a stereotaxic frame. A heating pad regulated by a temperature controller maintained the rat's rectal temperature between 36 and 39 °C. After opening the cisterna magna to prevent cortical swelling, a craniotomy was made over the region of the left frontal or parietal cortex to be studied. The dura was removed, and the exposed cortex was covered with warm saline or mineral oil to prevent drying.

2.2. *Stimulation procedures*

In 16 rats the electrode penetrations were made *perpendicular* to the cortical surface, and only one point per penetration was stimulated at a depth of 1.7 mm below the pia. This depth was selected since we[42] and others[15,17] had found minimum threshold values there, and it is clearly above the white matter (cf. Fig. 4B), except in the thinner cortex caudal to bregma (hindlimb region) where penetrations were made to a depth of 1.5 mm. The penetrations labeled 'A' in Fig. 1 illustrate several such perpendicular tracks. In general, the electrode penetrations were made 0.5 mm apart, surface blood vessels permitting; a total of 50–100 penetrations were made in each experiment. A Kopf 1760/1761 micromanipulator was used to position and advance a glass-insulated tungsten microelectrode (50–100 μm exposed tip) used for stimulating[39]. The electrode was connected to one terminal of a constant current unit (Grass PSIU6C) that relayed the output of a stimulator (Grass S88). The stimulus parameters were a 300 ms train of 0.25 ms cathodal (−) pulses at 350 Hz. In two animals, movements evoked by shorter trains (50–60 ms) were compared with movements evoked by the longer trains (300 ms); only the duration of the movements changed, not the movement produced. The maximum current used in these 'perpendicular' experiments was 100 μA. The currents delivered were monitored on an oscilloscope by measuring the voltage drop across a 10 kΩ resistor located in the return current path from a needle electrode inserted in the rat's tail to the other terminal of the constant current unit.

The remaining sixteen rats were studied using electrode penetrations made *parallel* to the dorsal (n = 14) or lateral (n = 2) surface of the cortex. In each of

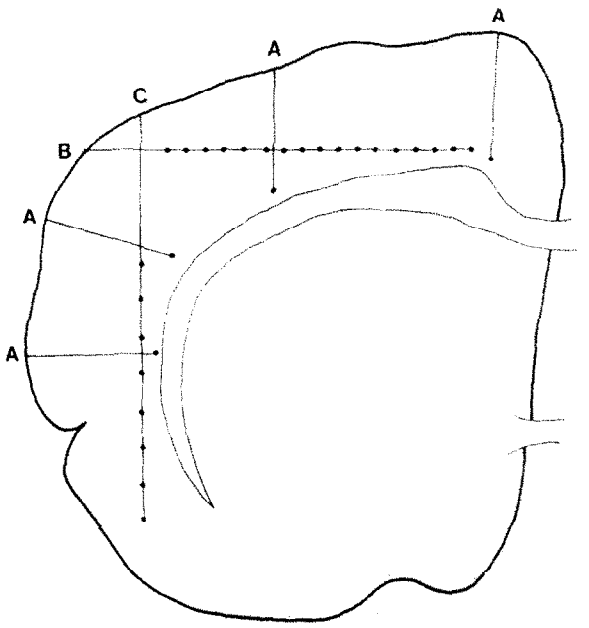


Fig. 1. Diagrammatic coronal section of rat hemisphere illustrating typical electrode penetrations perpendicular (A) to the cortical surface or parallel to its dorsal (B) or lateral (C) surface. Stimulation points are indicated by dots which are separated by 250 μm along B and 500 μm along C; dot at bottom of A is at 1.7 mm below cortical surface.

the 14 rats where penetrations were parallel to the dorsal surface (track 'B' in Fig. 1), 15–20 such penetrations were made rostro-caudally, spaced 250 μm (12 rats) or 500 μm apart (2 rats); the wider spacing allowed the entire dorsal motor cortex to be mapped. Stimulation was delivered every 250 μm along the 3–5 mm extent of each tract, with the result that 300–500 points were mapped per experiment. The stimulus parameters were the same as those used in the perpendicular experiments, with the exception that the maximum current limit was raised to 500 μA instead of 100 μA. The reason for raising the current intensity is that at the beginning of each of these parallel tracks the electrode tip was located in the superficial cortical layers where stimulation thresholds are much higher. Hall and Lindholm[19] found that thresholds in these superficial layers were between 400 and 500 μA and that the responses evoked by such high currents superficially were similar or identical to those evoked by lower currents deeper in the cortex. It has, however, been shown that such high currents are damaging to the cortex, particularly with the long trains we employed[2].

In the two rats where penetrations were parallel to the *lateral* surface of the cortex (track labeled 'C' in Fig. 1), several rows (7–9 tracks per row) of such penetrations were made, and the distance between penetrations was 0.5 mm. Each row began with a medial penetration and ended with a lateral penetration that extended from the dorsal cortical surface to the piriform cortex ventrally. Along each individual electrode track stimulation was delivered every 0.5 mm, except in the more medially located tracks perpendicular to the cortical surface where only one point was stimulated at a depth of 1.7 mm (as seen in track labeled 'A' in Fig. 1). In these two experiments the stimulation parameters were the same as previously described except that the maximum stimulation current used was only 50 μA.

Threshold current was defined as the lowest current reliably evoking the observed response. In determining threshold, the current was gradually reduced in successive trains until the movement ceased. The last effective current was designated as threshold, and 2 or 3 identical movement responses at this current were confirmed. Although it was not uncommon for the initial stimulating current to evoke several different movements, in general, at threshold current levels, only one movement was elicited from a given point. In the perpendicular penetrations and in the penetrations parallel to the lateral surface the initial stimulation current was 25–30 μA; if no response occurred, the current was increased until a response occurred or the current maximum (50 or 100 μA) was reached. In the penetrations parallel to the dorsal surface the initial stimulating current was 500 μA; once a response occurred at this or a lower value, this threshold current was used as the first current level at the next point. In all experiments manipulation or pinching of the limbs during threshold determination was done to lower the final threshold value[2]. At selected points in both perpendicular and parallel penetration experiments marking lesions were made (10 μA DC for 10 s) in order to facilitate subsequent histological reconstruction of the electrode tracks.

2.3. *Histology*

Following the conclusion of each experiment, the rat was perfused through the heart with saline (0.9%) followed by 10% buffered formalin. Brains were removed and placed in a 30% sucrose in formalin solution and allowed several days to sink. Frozen coronal, sagittal or horizontal sections were cut at 30–50 μm and stained with cresyl violet, thionin or a combined myelin-Nissl staining procedure using cyanine R and neutral red[11]. The sections from experiments using perpendicular penetrations (currents < 100 μA) usually showed no damage to the tissue. As expected, sections from the high current penetrations parallel to the dorsal surface showed damage; these tracks, especially their lateral portions, appeared as a series of small lesions (250 μm apart) strung together like beads on a string (cf. Fig. 9). The location of each stimulation point was determined on the histological sections to verify that it was in the gray matter; points located in the white matter were discarded. The stained sections were also used to determine the boundaries between the major cytoarchitectonic regions of the rat sensorimotor cortex, using the criteria described by several recent studies[15,62].

2.4. *Data analysis*

In order to generate a picture of the cortical pat-

tern of evoked movements, the locations of all gray matter stimulation points from representative experiments were plotted on drawings of the cortical surface as viewed from a dorsolateral and slightly rostral perspective. This drawing was created by making a large 3-dimensional reconstruction of the rat brain, based on the Nissl plates from the brain atlas of Paxinos and Watson[44]. A scaled 1 mm grid was traced onto the model, and a photograph was taken which was used to produce the line drawing used for the figures in this paper. Fig. 2 depicts this drawing with the 1 mm grid and cortical cytoarchitectural boundaries as determined on the Nissl plates of the atlas.

3. RESULTS

3.1. *Cortical cytoarchitecture*

The cytoarchitectonic boundaries in frontal and parietal cortex drawn on the surface of the hemisphere illustrated in Fig. 2 are in basic agreement with those drawn recently by Zilles et al.[62] and Donoghue and Wise[15]. Table I below shows the correspondence between the nomenclature used in Fig. 2 and that used by these two groups. Both groups described a narrow, parasagittal medial agranular area (AgM in Fig. 2), a lateral agranular area (AgL in Fig. 2), and a more lateral and caudal granular cortical area (Gr(SI) in Fig. 2). The only point of disagreement between these two studies was that Zilles et al. showed their AgM region extending straight rostrally toward the frontal pole while Donoghue and Wise claimed that their homologous area AgM wrapped around the frontal pole of the hemisphere and actually incorporated the far lateral portion of AgL on our map, a region which corresponds to Zilles' Prc3. Our map is in basic agreement with that of Donoghue and Wise on this point, and the agreement is based on our cytoarchitectonic analysis of the frontal cortex illustrated in the sagittal and horizontal Nissl plates in the Paxinos and Watson[44] rat atlas. These plates show AgM's pale layer III cortex extending laterally around the frontal pole, as Donoghue and Wise[15] have described. The insular or claustrocortex (Cl) is located far laterally on our map, just above the rhinal fissure, and includes Zilles' claustrocortex isocorticalis and his claustrocortex proisocorticalis, regions which correspond to Krettek and Price's[27] granular and agranular insular cortex. This region (Cl) is bordered dorsally at more caudal levels by the granular second somatosensory area, Gr(SII).

TABLE I

Terminology comparison

Fig. 2	*Donoghue and Wise*	*Zilles et al.*
AgM	Agm	Prcm
AgL	Agl	Prc1,2,3
Gr(SI)	SI	SmI
Gr(SII)	SII	SmII
Cl	–	Cli, Clpri

3.2. *Stimulation maps*

Since it was difficult to map in detail the entire 'motor' cortex in a single animal, most experiments concentrated on mapping responses in only one of the major subdivisions of the motor cortex: the forelimb area; the hindlimb area; the vibrissae area; and the jaw, lips and tongue area. The boundaries of each area were determined by finding non-responsive points outside the area or by finding points where the primary response for the experiment changed to a secondary response (e.g., where forelimb responses changed to hindlimb responses). Only one move-

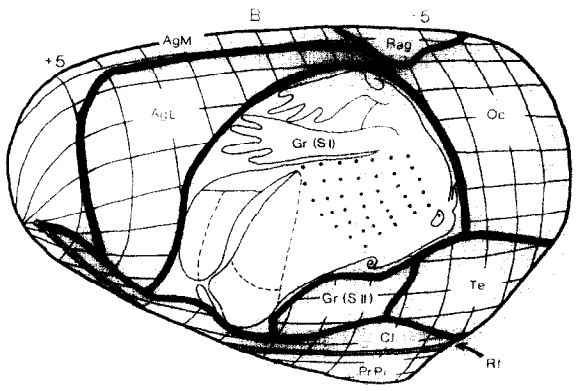


Fig. 2. Dorsolateral view of rat left hemisphere with 1 mm grid (boxes) and cytoarchitectonic borders (heavy lines) indicated (see text for fuller description of cytoarchitectonic parcellation). Frontal pole is at the left, midline is at the top, and numbers (+5, −5) denote distance in millimeters rostral and caudal to bregma. B, bregma; Rf, rhinal fissure; AgM, agranular medial; AgL, agranular lateral; Gr(SI), granular primary somatosensory; Gr(SII), granular secondary somatosensory; Cl, claustrocortex; PrPi, prepyriform; Rag, retrosplenial agranular; Oc, occipital; Te, temporal. A rattunculus has been superimposed on the Gr(SI) region after Welker[55].

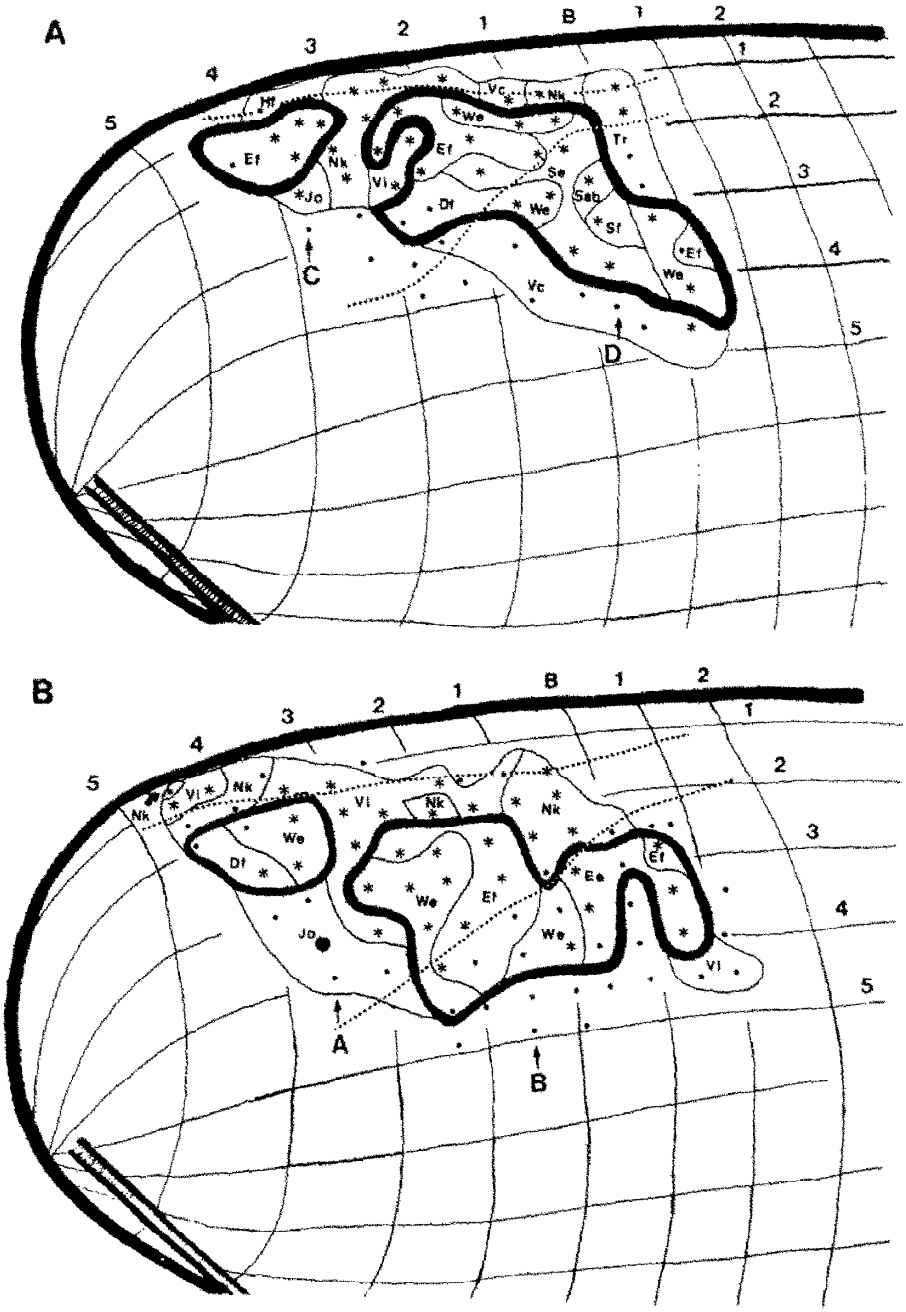


Fig. 3. Results of two (A, B) separate perpendicular penetration stimulation mapping experiments. In this and subsequent figures, frontal pole is at the left, midline is at the top, B denotes bregma, and numbers indicate distance in millimeters rostral or caudal from bregma or lateral from midline. Each square on grid represents 1 mm^2. Asterisks (*) denote stimulation points where threshold current was less than 25 μA, small dots (•) denote points where threshold current was less than 50 μA, and large dots (●) denote points where threshold current was less than 100 μA. Small dots outside areas enclosed by lines indicate points where no response was seen below maximum current used in that experiment. Dashed lines indicate cytoarchitectonic borders between AgM and AgL (medial line) and between AgL and Gr(SI) (lateral line). Movements evoked at a point are indicated by abbreviations (see Appendix). Large letters (A, B, C, D) refer to levels of coronal section plates in Fig. 4.

ment response is listed for each point, namely, the movement occurring at threshold current; at a few points, usually on the border between movement zones, two movements occurred at threshold, and in these cases, for simplicity, only one of the responses was listed on the map. Data from experiments that used relatively low current intensities (maximum of 50–100 μA to minimize the extent of current spread) are emphasized in this report because the smaller radius of current spread[1,33,47] is more likely to give an accurate picture of the pattern of motor cortical output. Data from experiments using higher current intensities are referred to *only* when the current intensities at the points discussed were low (less than 100 μA) or when the results agree with results from low current experiments.

3.2.1. *Forelimb maps*

Fig. 3 illustrates the results of two typical perpendicular mapping experiments that attempted to define the limits and internal organization of forelimb motor cortex. Note first that in each experiment *two* forelimb areas (outlined in heavy lines) were found, a large caudal area (CFL) beginning at bregma and a small rostral area (RFL) located near the frontal pole. Cortex where vibrissae (Fig. 3A) or neck (Fig. 3B) movements were evoked separated the two forelimb areas; the neck movements were observed as actual contractions of neck muscle groups seen in the surgical exposure used to drain the cisterna magna. Medial to both forelimb areas, a long parasagittal vibrissae and/or neck movement region was found. It was possible in some experiments (Fig. 3A) to find a rostral hindlimb representation (RHL) just medial to the RFL. This RHL was difficult to locate, even when it was expressly sought, and threshold currents here were usually greater than 40 μA. If a RHL was not found, neck (Fig. 3B), trunk or vibrissae movements were found in its place. Lateral to the RFL, jaw and/or tongue movements were evoked; lateral to the CFL, vibrissae movements were sometimes evoked (Fig. 3A). Histological analysis showed this region to be part of the vibrissae barrel field cytoarchitectonic area of Gr(SI). Caudal to the CFL, in many experiments a narrow zone of trunk movements (Fig. 3A) was found before the hindlimb representation began. Within the RFL, digit movements were usually evoked laterally while elbow movements were more often seen medially; shoulder movements were rarely seen in the RFL region. In contrast, in the CFL, digit movements were rarely evoked, but a laterally and caudally located shoulder movement zone was often found, with elbow and wrist movements found more medially (cf. Fig. 3A). Note also that the points where a particular movement was elicited were usually clustered together, forming an irregularly shaped patch or zone. This last observation held true for all regions of the motor cortex, as will be seen below.

Fig. 4 illustrates Nissl-stained coronal sections from the corresponding levels marked A–D in Fig. 3. The sections through the RFL (Fig. 4A, C) clearly show that the RFL is located in AgL; however, the RHL region (Fig. 4C, Hf = hip flexion) appears to extend into AgM. Sections through the CFL (Fig. 4B, D) show that the CFL lies in both AgL and Gr(SI) cortex. Thresholds usually increased as the penetrations moved deeper into Gr(SI) cortex (cf. Fig. 4B) but, in some cases, low thresholds were found throughout Gr(SI) as seen in Fig. 4D. Note, however, that in Fig. 4D thresholds were lower in the more medial penetrations which were located in AgL, and thus the generalization that ICMS thresholds are higher in Gr(SI) cortex than in AgL still seems to hold[15]. The boundaries between AgM, AgL and Gr(SI) are also indicated in Fig. 3 by the dashed lines.

In the experiments with penetrations parallel to the dorsal cortical surface the same basic observations were made, but a more detailed map could be constructed because of the high density of stimulated points. Fig. 5 illustrates the results of one experiment in which cortex rostral to bregma was mapped. Two forelimb areas are again seen, with a zone of neck and vibrissae movements between them. Note that within the CFL supination movements were evoked caudally and laterally while pronation movements were evoked caudally and medially. These movements were also found in these locations in other parallel experiments and in some low current perpendicular experiments as well. Just rostral to the RFL a small rhinarium (nose twitch) region was seen; this also was occasionally seen in perpendicular experiments where it was located either rostral or caudal to the RFL. Note also the medially located vibrissae motor region and the caudally located narrow zone of

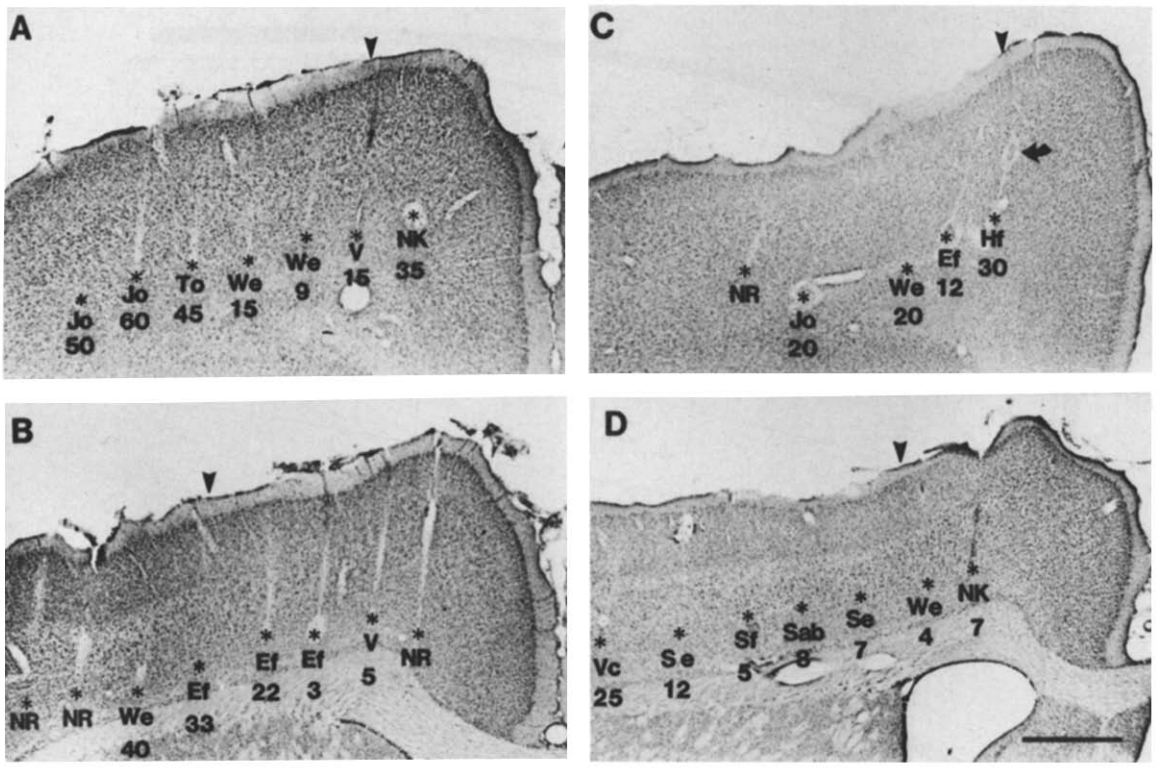


Fig. 4. Nissl-stained sections from perpendicular mapping experiments shown in Fig. 3 and labeled A, B, C and D in that figure. Numbers below movement abbreviation in this figure indicate threshold current in μA at that point. A: section through RFL area with jaw and tongue movements laterally and vibrissae and neck movements medially. B: section through CFL with vibrissae movement zone medially and some responsive points in Gr(SI) laterally (border between AgL and Gr(SI) indicated by arrow on cortical surface). C: section through RFL and RHL, with jaw movements located laterally. D: section through CFL with neck movements medially. Arrow on cortical surface indicates border between AgL and Gr(SI); note that even though threshold currents are low in Gr(SI), they are still higher than those medially in AgL. Scale bar in lower right of D indicates 1 mm for entire figure.

trunk movements separating the CFL from the hindlimb motor region. Trunk movements usually consisted in a slight curvature of the back away from the stimulated side of the animal.

3.2.2. *Hindlimb maps*

Fig. 6 illustrates the results of two perpendicular mapping experiments that focused on the hindlimb motor cortex (HL). This region usually began at bregma and extended 2 or 3 mm further caudally. Medially, the HL area was bordered by cortex where vibrissae movements or eyelid movements were elicited. Rostrally, a zone of trunk movements separated the HL area from the CFL region. Laterally, a caudal

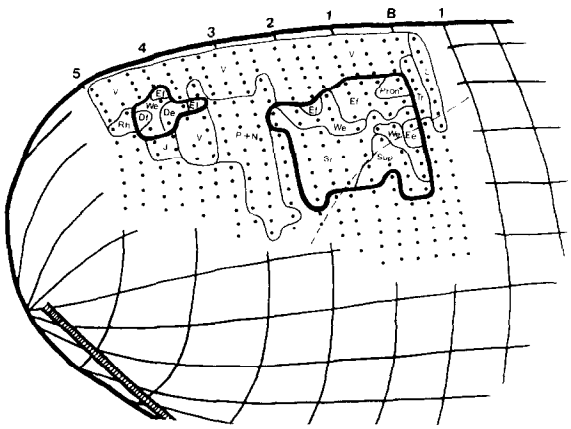


Fig. 5. Results of stimulation mapping experiment employing electrode penetrations parallel to dorsal cortical surface. Note much higher density of stimulated points achieved by using this approach. The two forelimb areas (RFL and CFL) are outlined by heavy lines. Dashed line running through CFL indicates border between AgL medially and Gr(SI) laterally. Note narrow region where trunk movements were elicited caudal to CFL and just rostral to leg (L) region.

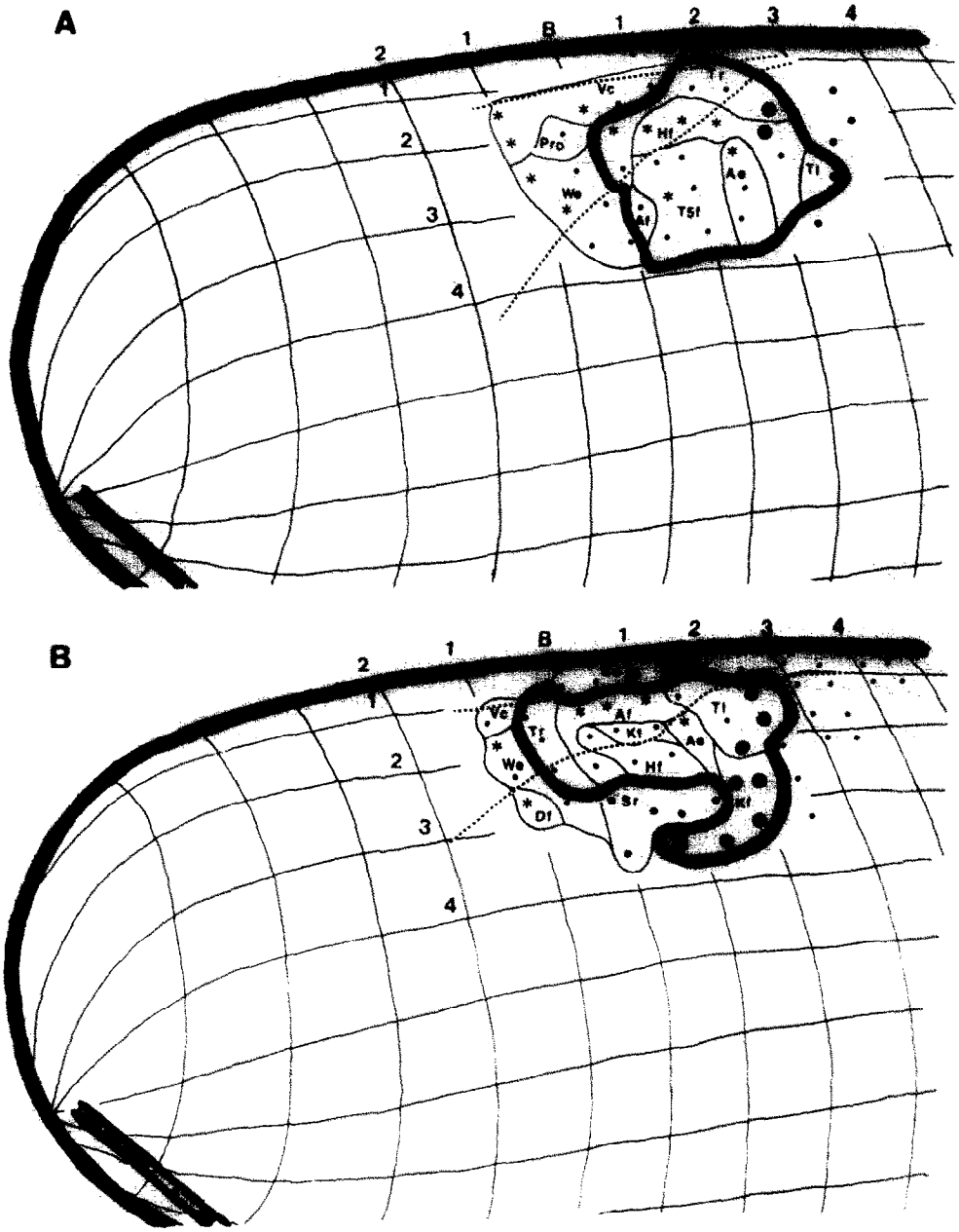


Fig. 6. Results of two separate (A, B) stimulation mapping experiments of hindlimb area (heavy line) using perpendicular penetrations. A: note trunk region medially and tail region caudally. B: note eyelid responses medially.

extension of the CFL usually bordered the HL cortex, as is seen in Fig. 6B. Caudally, a zone where tail movements were evoked was usually found. Within the HL area, hip movements were usually found most medially (Fig. 5A), while knee, ankle and toe movements were located more laterally. Note that the HL motor cortex was located in both the Gr(SI) and AgL cytoarchitectonic regions but did not appear to extend into AgM. This same basic pattern was also observed in experiments with penetrations parallel to the dorsal cortical surface, and an example is illustrated in Fig. 7.

3.2.3. *Vibrissae maps*

Fig. 8 illustrates the results of two perpendicular experiments which mapped the vibrissae motor cor-

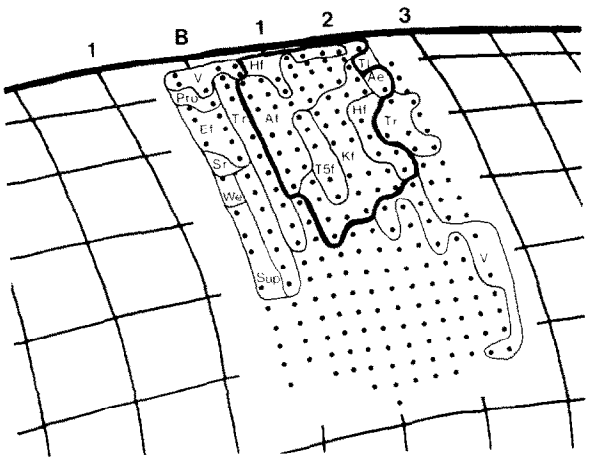


Fig. 7. Results of one stimulating mapping experiment of the hindlimb region using penetrations parallel to the dorsal cortical surface.

tex. This region is located most medially in the motor cortex and forms a long, narrow, parasagittal strip of cortex extending from near the frontal pole to 1 or 2 mm behind bregma. Laterally, it is bordered by hindlimb, trunk, forelimb and neck motor cortical areas. Medially, in some experiments (Fig. 8B) it was possible to find unresponsive points which formed the basis for delineating the medial border. The most interesting aspect of the vibrissae area's organization was that there seemed to be largely separate zones for ipsilateral and contralateral vibrissae movements, although along the borders between these zones bilateral whisker movements could be evoked. There was no evidence for any type of 'whisker-by-whisker' representation; rather, throughout this region vibrissae movements involving clusters of at least 3–5 of the whiskers were elicited. The majority of effective points for evoking whisker movements at low thresholds appeared to be located along the border between AgM and AgL. In the experiments with penetrations parallel to the dorsal cortical surface the low threshold vibrissae movements appeared to begin at the AgM-AgL border, and most vibrissae movements were evoked from within AgM, as is illustrated in Fig. 9.

3.2.4. *Jaw, lips and tongue maps*

Fig. 10 depicts the results of two perpendicular experiments that were devoted to mapping the jaw, lips and tongue area (JLT) of the motor cortex. It is clear from the maps that this region was very large relative to the other subdivisions of the motor cortex described above. Rostrally and laterally the JLT was surrounded by unresponsive points, while medially and caudally its border was defined by points where vibrissae, neck or forelimb movements were evoked. The jaw movement responses usually consisted of jaw opening, and such points were most often located just lateral to the RFL as is seen in Fig. 10A. Note that in this same map there was a second jaw region located more laterally, at the rostral border of the JLT region. In the other map (Fig. 10B) there was only a single jaw point. The lower lip responses consisted of raising the contralateral lower lip near the angle of the jaw. This response was best seen if the lower lip was pulled down slightly just prior to stimulation. This lower lip region varied in size but was always found as part of the caudal border of the JLT region. Tongue movement responses were of two basic types, either strong tongue protrusions (To = tongue out) or much weaker twitches of restricted parts of the tongue such as the tip (Ttip), the dorsum (Td), the ventral surface (Tv), or the right (contralateral to stimulation) side (Tr = tongue right). In general, the strong tongue out responses were located medially and rostrally in the JLT region, and the weaker tongue twitch responses were found laterally.

The two experiments with long perpendicular penetrations parallel to the lateral cortical surface revealed that the lateral border of the JLT region actually extended down to the rhinal sulcus. Fig. 11 depicts a series of electrode tracks from one of these experiments. Note at the most rostral level (Fig. 11A) that the orientation of the more lateral radial cell columns (dashed lines) is such that these columns extend to the cortical surface near the rhinal sulcus (area C1); a 'perpendicular' penetration (marked with * in Fig. 11A) would cross these columns where jaw and tongue movements were evoked by ICMS (Jo and T) and stimulate a non-responsive region. This probably explains the far lateral non-responsive points in Fig. 10. Fig. 11 also illustrates several other similar coronal sections from more caudal levels which show that the more lateral tongue twitch responsive points were also located in the C1 cortex near the rhinal sulcus. These same sections also demonstrate that ICMS was effective in evoking movements from the granular sensory cortex, especially at

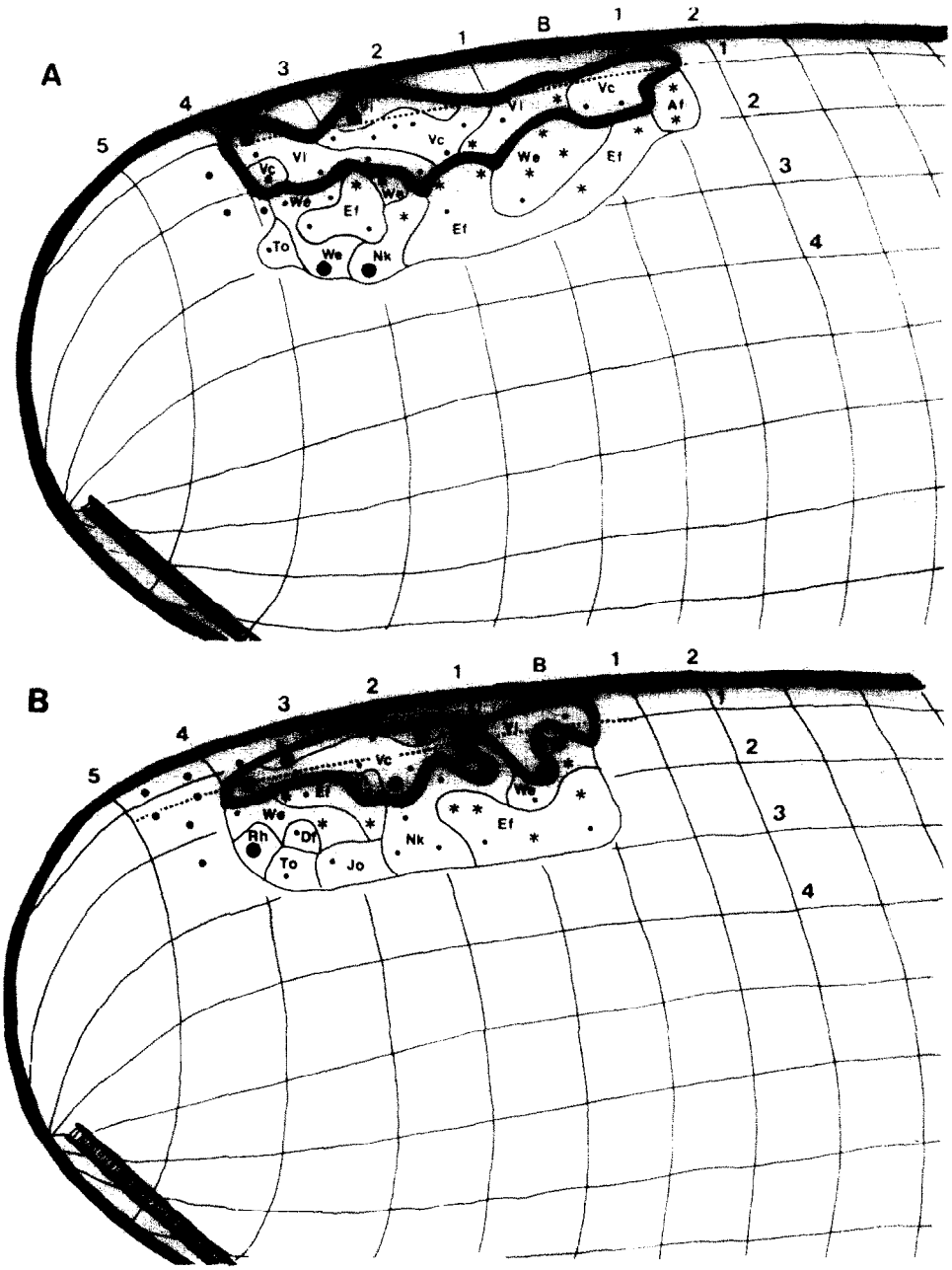


Fig. 8. Results of two separate (A, B) stimulation mapping experiments of the vibrissae area using perpendicular penetrations. Note that there appear to be separate regions for contralateral and ipsilateral vibrissae movements. Dashed line indicates border between AgM and AgL.



←

Fig. 9. Coronal section from experiment employing penetrations parallel to dorsal cortical surface. Note that the cytoarchitectonic border between AgL and AgM appears to correspond to the transition from CFL to vibrissae area motor cortex. Thresholds in μA from medial to lateral, beginning with most medial vibrissae (V) point, are 50, 35, 10, 80, 100, 180, 275, 240, 190, 190, 190, 340 and 500; remaining far lateral points produced no movements with currents of 500 μA.

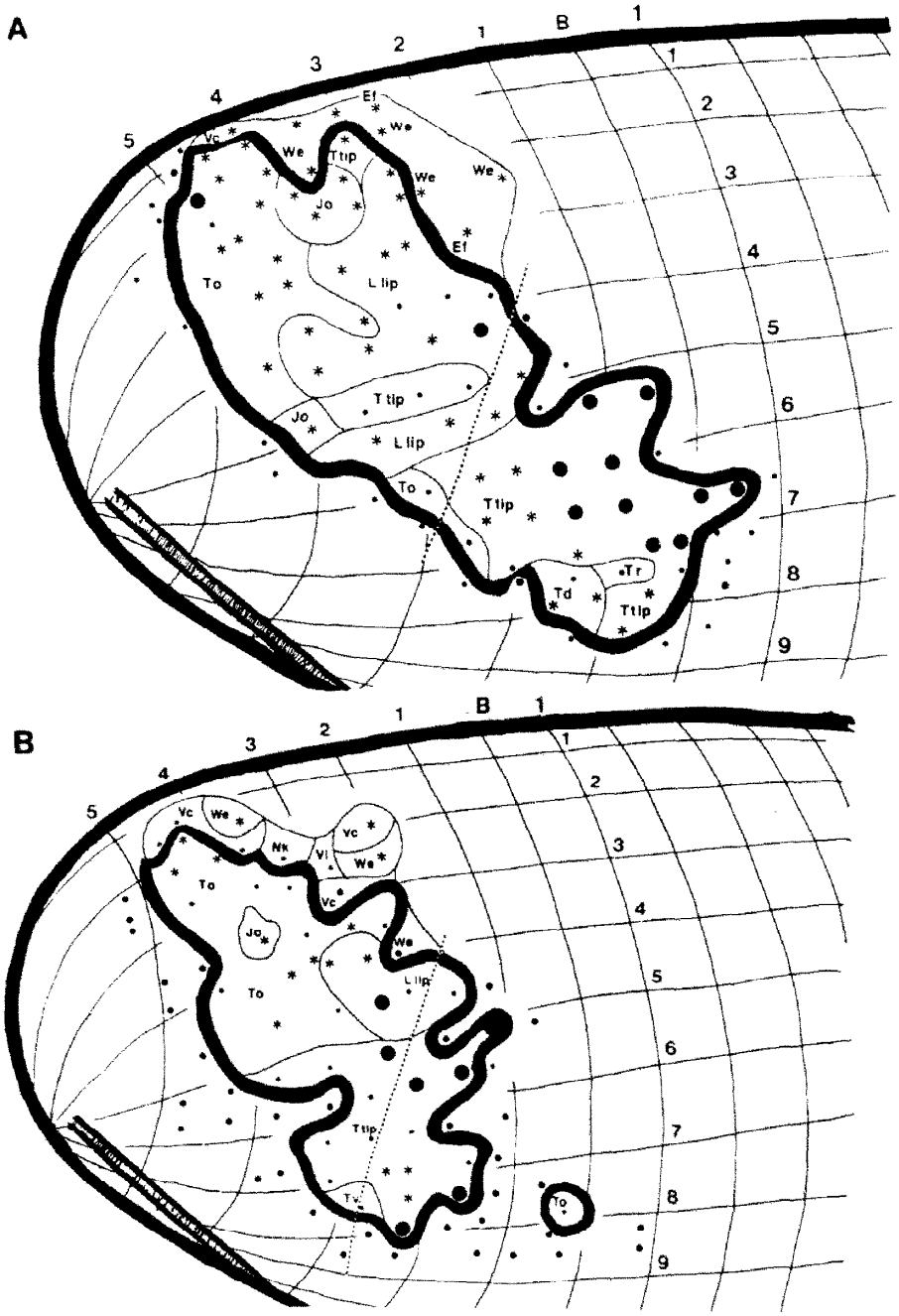


Fig. 10. Results of two separate (A, B) stimulation mapping experiments of the jaw, lips and tongue area using perpendicular penetrations. Dashed line indicates the border between AgM and Gr(SI). Note the significant overlap of the motor area with the granular sensory cortex.

rostral levels where the granular cortex borders the AgL cortex (Fig. 11B). The data from the coronal sections in Fig. 11 is shown projected onto the cortical surface in Fig. 12.

4. DISCUSSION

4.1. *Microstimulation*

The primary method used to create the motor

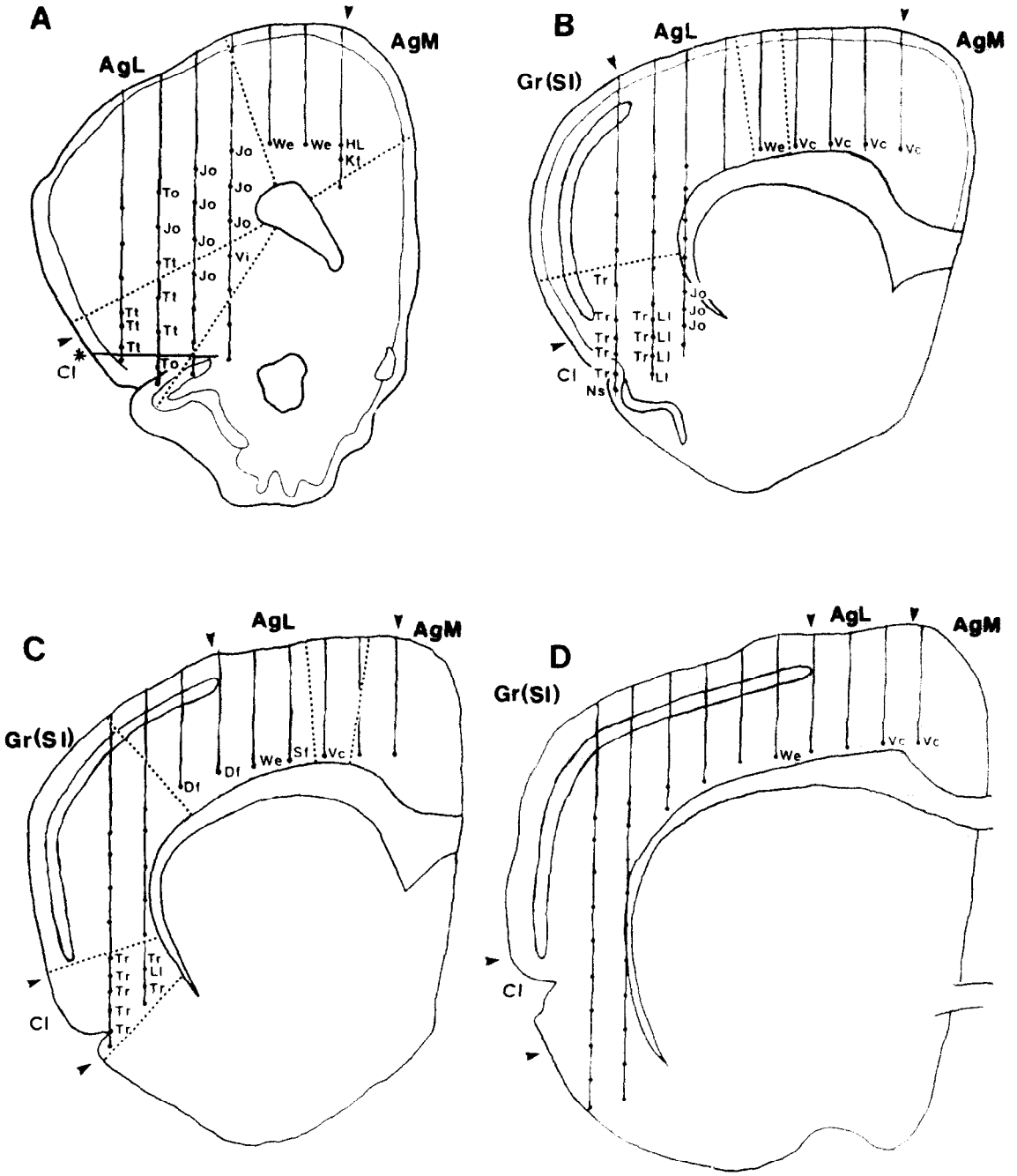


Fig. 11. Line drawings of 4 coronal sections from an experiment which mapped the jaw, lips and tongue region using penetrations parallel to the lateral cortical surface. Cytoarchitectonic boundaries are indicated by arrowpoints and labels on cortical surface; granular cortex is also indicated by outlined layer IV. A: coronal section through RFL-RHL region. Note extension of responsive JLT cortex into Cl. Also note how a typical 'perpendicular' track (heavy line marked with *) would stimulate a point located in a non-responsive portion of the cortex. B: coronal section through CFL. Again note ventral extension of JLT cortex almost into piriform cortex. C: coronal section through CFL. Note that at this level most of Gr(SI) is unresponsive, but that tongue and lip responses begin again in Cl. D: coronal section through CFL. Note that at this more caudal level Gr(SI) and Cl are both non-responsive, except for one point at border with AgL. Stimulation points and electrode tracks are separated by 500 μm.

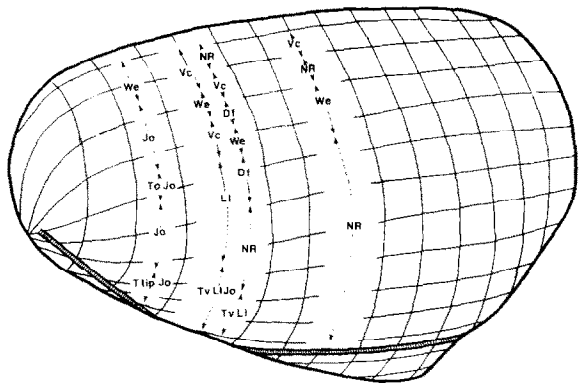


Fig. 12. Projection of primary movement responses from 4 sections illustrated in Fig. 11 onto cortical surface to emphasize ventral extension of JLT motor area down to rhinal sulcus.

maps of cortical organization in this study was intracortical microstimulation (ICMS), and over 85% of the responsive points had current thresholds less than 50 μA. It has been estimated that the effective radius of current spread in the cortex of a 50 μA stimulus current (0.2 ms duration) is approximately 500 μm[33,47]; this distance represents the margin of error associated with the 'borders' drawn on the maps in the present study. ICMS is, of course, an artificial, non-physiological method of activating the cortex, and its results must be interpreted cautiously due to some uncertainty concerning the effective spread of its direct and indirect effects[1,24]. Nonetheless, the responses evoked by low intensity ICMS do appear to reflect the actual functional pattern of cortical output. The recent study of the primate motor cortex by Cheney and Fetz[10] has demonstrated a remarkable consistency between corticomotoneuronal cell spike-triggered and single pulse ICMS stimulus-triggered patterns of muscle facilitation. (Since their study did not employ trains of stimulation they were able to avoid the summation effects associated with repetitive stimulation.) On the basis of their results they have suggested that the output of primate precentral cortex is organized into small clusters of corticomotoneuronal cells (0.5 mm in diameter) that facilitate a common group of muscles; other laboratories have proposed a similar organization for the primate motor cortex[3,30,38]. This pattern is consistent with that seen in the present study where small, irregularly shaped 'movement zones' were defined by ICMS and suggests that our results are an approximation to the actual functional output organization of the rat's motor cortex. (The major alternative to ICMS, namely, cortical surface stimulation, requires much higher intensities of stimulating current (400–500 μA minimum for mapping[25]). At these high current levels the large effective radius of current spread from the stimulating electrode (1.1–1.5 mm on the cortical surface[47]) precludes the construction of the 'fine grain' maps of cortical organization that were the objective of the present study. Underestimation of this radius of current spread in several surface stimulation studies[25,45] can lead to the erroneous impression that, as a rule, even the cortical representations of antagonistic muscles or muscle groups overlap almost completely. This criticism is not meant to imply that there is *no* overlap of cortical projections to even antagonistic muscle groups but to correct the impression from these surface stimulation studies that overlap is the rule rather than one of several cortical 'strategies' used to represent movements[3,23]. It is interesting that the results of our experiments where intracortical stimulation with currents up to 500 μA in the more superficial cortical laminae was employed agree so well with the results seen when only currents below 50–100 μA were used. This observation illustrates the critical importance of the *depth* in the cortex at which the stimulation is delivered and also confirms Hall and Lindholm's observation that the *depth* at which stimulation is delivered changes the threshold for evoking a movement but not the movement itself[19].

4.2. *Motor output organization*

Fig. 13 represents a composite motor output map of rat frontal and parietal cortex, summarizing the pattern of motor responses evoked at low threshold currents (<50 μA) from all the experiments. Due to the variability between the maps of individual animals described above, this map certainly represents an idealized picture, but it does illustrate 'typical' features and provides a convenient means of discussing our results and comparing them to others' findings. In Fig. 13 the 1 mm^2 grid and cytoarchitectonic borders (heavy solid lines) are as described in Fig. 2. The heavy dotted lines outline the major subdivisions of motor cortex such as the forelimb or hindlimb areas; the light dotted lines indicate the

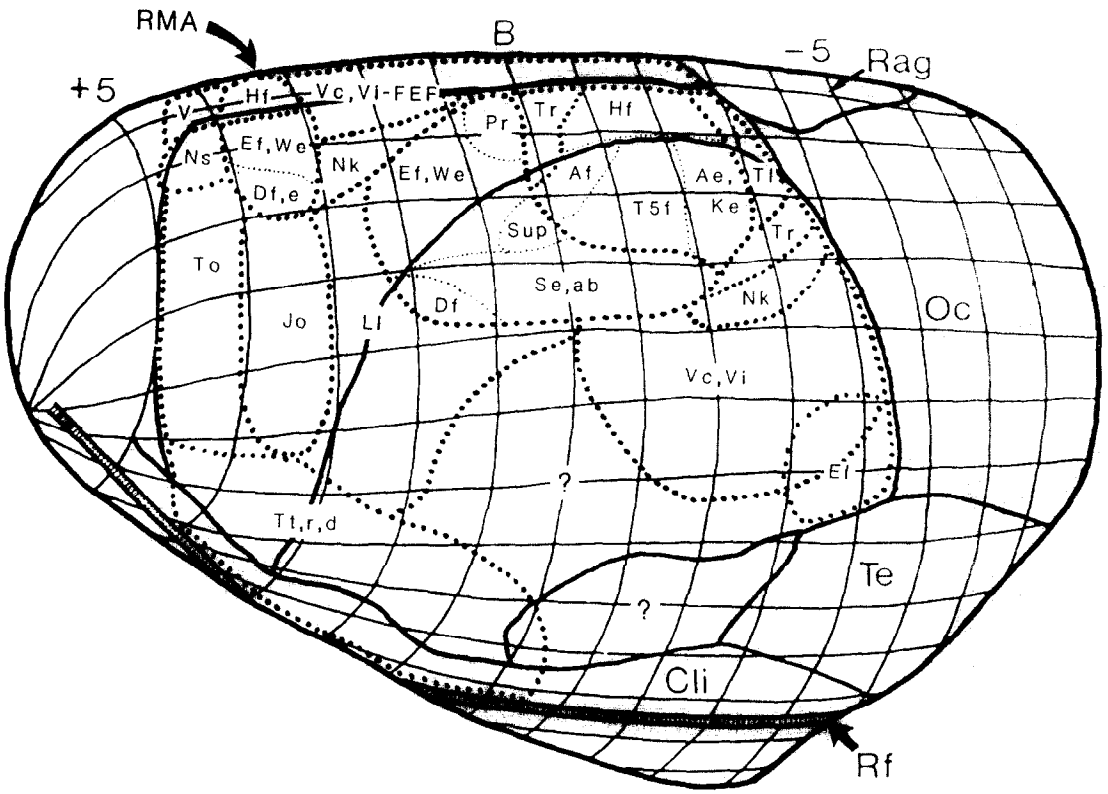


Fig. 13. Idealized summary map of responses evoked with currents below 50 μA. The light lines denote a 1 mm^2 grid, and the heavier solid lines indicate the cytoarchitectonic borders as described in Fig. 2. The heavy dotted lines surround the major subdivisions of the motor cortex, and the light dotted lines indicate the internal organization within the forelimb and hindlimb areas. The arrow (RMA) indicates the rostral motor area. Note the correspondence between the cytoarchitectonic border between AgM and AgL (solid line about 1 mm lateral to midline) and the border between vibrissae motor-frontal eye field region (Vc, Vi-FEF) and the somatic motor cortex. See text for fuller description. Rf, rhinal fissure; Rag, retrosplenial agranular cortex; Oc, occipital (visual) cortex; Te, temporal (auditory cortex); Cli, claustral (insular) isocortex; B, bregma; ?, parts of Gr(SI) and Gr(SII) that may also be motor regions.

'typical' pattern of the internal organization of these regions. (Dotted lines are used to denote all these borders partly to suggest that these borders are not fixed or invariant among different animals.) The relative location of the major body part subdivisions of motor cortex illustrated in Fig. 13 agrees substantially with the maps generated by other studies[17,19,49]; differences between our results and these other studies will be discussed below. Both the rostral (RMA, arrow) and caudal motor areas appear to possess a rough somatotopic organization, in agreement with previous studies[17,19,49,50,60], but it should be noted that many examples can be found where adjacent regions of cortex do not 'represent' adjacent body parts; e.g., the caudal hindlimb area and the vibrissae area border each other. The cytoarchitectonic border between AgM and AgL (indicated by the heavy solid line running parasagitally about 1 mm lateral to the midline and then curving laterally near the frontal pole) appears to roughly correspond to the functional border between the vibrissae-eye movement region medially and the rest of the motor area laterally. In contrast, the more lateral border between AgL and Gr(SI) does *not* appear to correspond to a functional border as far as motor output is concerned.

The extent of cortex devoted to the various movements evoked by microstimulation varied but often consisted of a patch of cortex 1–3 mm^2 in area. Several such distinct movement zones could also be defined *within* the forelimb or hindlimb areas. This finding of an internal organization to these regions is at odds with the recent report of Gioanni and Lamarche[17] who found that EMG activity could be evoked in 4–6 different forelimb or hindlimb muscles from virtually the entire forelimb or hindlimb region;

they concluded that the rat limb representations lack obvious internal organization. Although our study did not employ EMG recordings, Hummelsheim and Weisendanger[22] have recently reported that gastronemius EMG activity can be evoked by microstimulation ($<30\,\mu A$) of only a small region of rat hindlimb motor cortex and not the entire hindlimb region; remarkably, their effective region corresponds almost exactly with the small ankle flexion (Af) region in Fig. 13. This correspondence supports our finding of some internal organization of the rat's limb representations into separate movement zones or patches, similar to that seen in primates[10,18,30,38,40,53].

The approximate area of the cortex devoted to the major somatotopic subdivisions of the rat motor cortex is listed below in Table II.

4.3. *Overlap of motor and sensory cortex*

The map in Fig. 13 illustrates that most of the primary sensory cortex in Gr(SI)[9,55] can also be considered to be motor cortex, including the representations of the hindlimb, forelimb, lower lip, tongue and vibrissae. Previous studies in the rat have also found motor-sensory overlap but have noted differences in the amount of overlap for various body parts. Specifically, overlap was reported to be complete for the hindlimb representations, partial for the forelimb representations and non-existent for the face and head representations[14,19].

Concerning the hindlimb representations, our map shows a distinct, medially located portion of the hindlimb motor area in AgL that does not overlap with the sensory hindlimb granule cell field in Gr(SI). This medial hindlimb region in AgL does not respond to peripheral sensory inputs in the anesthetized rat and also contains corticospinal neurons which project to the lumbar enlargement[51], data which indicate that the overlap between sensory and motor hindlimb cortical areas may not be complete. Donoghue et al.[14] described the rat's hindlimb area as a region of sensorimotor 'overlap' cortex, defined as cortex which received projections from both the ventrobasal and ventrolateral thalamic nuclei. However, their tracer injections may have involved both the agranular and granular portions of the hindlimb region defined by the present study. The resolution of this question will depend on the results of anatomical studies of thalamocortical projections employing extremely small injections of two separate tracers into the AgL and Gr(SI) regions of the hindlimb area within the same animal.

TABLE II

Approximate area of cortex devoted to movements of various body parts

Vibrissae	17 mm^2	(V – FEF = 8 mm^2 + V – Gr(SI) = 9 mm^2)
Tongue	16 mm^2	(To = 4 mm^2 + Tt,r,d = 12 mm^2)
Forelimb	12 mm^2	(CFL = 10 mm^2 + RFL = 2 mm^2)
Hindlimb	8 mm^2	(CHL = 7 mm^2 + RHL = 1 mm^2)
Lower lip	5 mm^2	
Jaw	3 mm^2	

Concerning the forelimb representations, the caudal forelimb area corresponds closely with the forelimb area described by earlier mapping studies of the rat motor cortex[17,19,49,60] and is located in both the AgL and Gr(SI) cytoarchitectonic fields. The responsive cortex in Gr(SI) in Fig. 13 is shown as coextensive with the forelimb granule cell patch but in individual experiments may not always include its more lateral region, probably due to the greater sensitivity of motor responses in Gr(SI) to the level of anesthesia[17]. This last observation is a likely reason why previous studies[15,19] reported only partial overlap of forelimb motor and forelimb sensory cortex. The caudal forelimb motor area outlined in Fig. 13 coincides almost perfectly with the caudal patch of cortex which gives rise to corticospinal projections to the cervical enlargement[51,57]. The importance of the entire caudal forelimb region (both the AgL and Gr(SI) components) for motor control of the forelimb in the rat was first demonstrated by Castro[7] who found that small, bilateral motor cortex lesions that appear to be largely confined to AgL produced forelimb use deficits which showed significant recovery 1–2 weeks after the lesions; in contrast, rats with large lesions which appear to have involved both the AgL and Gr(SI) portions of the caudal forelimb area showed little recovery from their forelimb use deficits even 3 months after the lesions. Similar results have been seen more recently by Price and Fowler[46].

Concerning the lower lip and tongue representations, our map is the first indication that there might be overlap of the motor and sensory cortical regions for these body parts. This zone of overlap not only includes the Gr(SI) somatosensory representation of

the lower lip and tongue but also appears to include the visceral sensory taste representation located in the insular (Cl) cortex[4,43,61]; the motor responses evoked from this taste sensory cortex seem to consist largely in the small, tongue twitch type of responses. The previous lack of reports of motor responses from these cortical regions is due largely to the absence of mapping studies of the motor representation of these body parts in the rat; in contrast, the sensory representation of the head, face and tongue has been studied in detail in the rat[9,55,61]. In addition, reports of a lack of overlap between the motor and sensory head representations[15,19] have been primarily referring to the lack of overlap between the motor and sensory *vibrissae* representations (not the jaw, lips and tongue representations). The importance of the tongue motor representation in Gr(SI) was demonstrated by Castro[8] who found that bilateral lesions of this area seriously impaired the ability of the rats to protrude their tongue and lick up rat food pellets; these deficits lasted as long as the animals were tested (up to 6 months).

The area occupied by the motor vibrissae and eyelid area in Gr(SI) is depicted on the map following the description by Gioanni and Lamarche[17] of their posterolateral vibrissae motor area; comparison of this region in Fig. 13 and the same region in Fig. 2 reveals that this area corresponds to the vibrissae barrel field of Gr(SI)[9,55]. In our own experiments we also evoked vibrissae movements from this region but did not study it systematically because the responses were somewhat inconsistent between animals. This is also likely due to the reported greater sensitivity of these vibrissae motor responses from Gr(SI) to the depth of anesthesia[17]. The question marks in the remainder of Gr(SI) and in Gr(SII) are meant to indicate that these areas may also be motor areas, but have not yet been studied carefully in the rat. The overall pattern that emerges from these findings is that the entire somatosensory cortex of the rat may also be described as a motor region, similar to the pattern in the primate[36,56,59]. This 'motor' characteristic of 'sensory' cortex supports Diamond's[13] proposal that layer V of the entire cortex should be termed the motor cortex, due to its substantial direct or indirect projections to motor neurons.

4.4. *Vibrissae motor area and frontal eye fields*

The map in Fig. 13 locates the primary motor representation of the vibrissae along the midline, in agreement with previous studies[15,17,19,49]. However, this region should *not* be considered as part of the primary motor area, as these previous studies have classified it, but rather should be considered as a portion of the rat's frontal eye field region. The rationale for this 'change of status' is based on the following discussion. Most of the vibrissae area is found within AgM, and the vibrissae motor area's lateral border appears to largely coincide with the cytoarchitectonic border between AgM and AgL, as was illustrated in Fig. 9. We and others[15,17,19,49,52] have found that stimulation in AgM also produces eye, eyelid or head movements, as well as pupillary constriction, and it has been suggested that this region corresponds to the rat's frontal eye field[15,19]. (Experiments now underway in our laboratory by Robert Frysztak have demonstrated that microstimulation with currents as low as $10\,\mu$A of a single point in the vibrissae area can elicit contralaterally directed eye movements along with bilateral vibrissae movements and pupillary constriction.) Since AgM has a strong projection to the superior colliculus[20,31,41], and since electrical stimulation of the superior colliculus has recently been shown to produce *bilateral* vibrissae movements along with eye and head movements at low thresholds[34], it is plausible to assume that the simultaneous vibrissae, eye and head movements elicited from AgM are due to activation of the projection from AgM to the superior colliculus. It thus appears that the border between AgL and AgM is a major functional division in the cortex, separating somatic sensorimotor cortex from eye-head orientation cortex. The more lateral somatic sensorimotor cortex (AgL, Gr(SI), Gr(SII)) possesses corticospinal projections[15,21,51,57] while the more medial frontal eye field cortex (AgM and anterior cingulate (AC)) possesses strong projections to the superior colliculus[20,31,41] but lacks corticospinal projections[51]. In addition, AgM and AC have significant reciprocal connections with the visual cortex[35]. Lesion studies have proposed that the anteromedial frontal cortex, including our vibrissae area, functions as part of a 'spatial mapping system'[26] which 'includes the superior colliculus and serves attentional/orienting functions'[12]; rats with le-

sions of this cortex fail to learn the Morris water task and are impaired at learning the location of reward in the radial maze task[26], as well as showing significant contralateral neglect[12]. It should be pointed out that near the frontal pole the border between AgL and AgM may lose some of its significance as a border between somatic and eye-head functional regions since the RFL and RHL are located in AgL *and* AgM and both give rise to corticospinal projections[15,51].

Within the vibrissae area, we consistently found largely separate zones where either contralateral or ipsilateral vibrissae movements could be evoked as well as regions where bilateral vibrissae movements were elicited. Ipsilateral vibrissae movements were only rarely seen by Sanderson et al.[49] in their stimulation study of the rat motor cortex. The reason for this disagreement is not clear but may lie in the different stimulation parameters used in each study. In support of our finding of both contralateral and ipsilateral vibrissae movements, Gioanni and Lamarche[17] reported that bilateral vibrissae movements could be evoked from their anteromedial vibrissae motor region. In addition, unilateral stimulation of the superior colliculus produces bilateral vibrissae movements in the anesthetized rat[34]. The bilateral vibrissae movements seen after unilateral cortical stimulation are also reminiscent of the bilateral control of facial musculature by each side of the cortex seen in man[28].

One study[17] has reported a 'vegetative zone' along the midline of the rat motor cortex where microstimulation elicits pupillary constriction responses; we have also seen such responses in some of our current mapping experiments (see above), but have found that this zone largely overlaps with the vibrissae motor area and eye movement region rather than existing as a separate entity.

4.5. *The rostral motor area: the rat's supplementary motor area?*

The second, more rostral motor area (RMA, arrow in Fig. 13) was located near the frontal pole and could always be distinguished from the caudal primary motor area by an intervening strip of cortex where neck or vibrissae movements were evoked. Within the rostral motor area the RFL appeared to lie within the AgL cytoarchitectonic region and overlapped with the rostral patch of corticospinal neurons projecting to the cervical enlargement[42]. The existence of the RFL has recently been confirmed by Sanderson et al.[49]. Although Woolsey and his coworkers[60] did not specifically describe a rostral forelimb motor area, close inspection of his figurine map (see his Fig. 122) does reveal a point in rostral frontal cortex where surface stimulation produced elbow flexion. This point was separated from his primary forelimb region by cortex where neck or vibrissae movements were elicited. Hall and Lindholm[19] did not describe a rostral forelimb area, perhaps because of a small gap in the points they mapped in the frontal cortex located at about 3.5 mm anterior to bregma and 1.5 mm lateral to the midline (cf. their Fig. 4). The failure of the recent study by Gioanni and Lamarche[17] to find a rostral motor area is surprising; however, their single forelimb motor area extends rostrally to include the region we have designated as the RMA. In addition, their study did find that neck or vibrissae movements could be elicited from the cortex between our two motor areas, but they reported that these responses were always accompanied by forelimb movements as well. In our mapping of this narrow region, neck or vibrissae movements always had lower thresholds than any forelimb responses that could also be evoked here. Further evidence in favor of the existence of a separate RMA is the finding of a second, small, rostral hindlimb region (RHL) just medial to the RFL area. The RHL appeared to be located in AgM and overlaps with a small rostral patch of corticospinal neurons projecting to the lumbar enlargement[51]. Only hip flexion and knee flexion movements were evoked from the RHL, and the thresholds were high (>40 μA), perhaps due to the small number (10–25) of corticospinal neurons found here[51]. Sievert[51] has also reported a corticospinal projection from this region to the thoracic cord, located in between and partially overlapping the projections to the cervical and lumbar enlargements; this observation is consistent with the occasional finding of trunk movements medial to the RFL.

The existence of both a RFL and a RHL suggests that the rostral motor area of the rat may correspond to the supplementary motor area (SMA) of the monkey, since the SMA contains representations of both the forelimb and hindlimb[32,58,59] which give rise to corticospinal projections to cervical and lumbar regions of the spinal cord[37]. Further support for this

suggestion comes from the finding that this rostral motor area of the rat is much less responsive to peripheral sensory inputs than the primary motor area[51], a pattern also seen when the sensory inputs to the monkey SMA and primary motor area are compared[5,58]. The lack of peripheral sensory input to the rostral motor area also makes it unlikely that this region is homologous to the monkey's premotor area (PM) since, although PM also gives rise to corticospinal projections[54], more than 50% of PM neurons respond to peripheral somatosensory inputs[48].

5. SUMMARY

In conclusion, the rat primary motor cortex appears to be organized into irregularly shaped patches of cortex devoted to particular movements. The location of major subdivisions such as the forelimb or hindlimb areas is somatotopic and is consistent from animal to animal, but the internal organization of the pattern of movements represented within major subdivisions varies significantly between animals. The motor cortex includes both agranular primary motor cortex (AgL) and, in addition, a significant amount of the bordering granular somatic sensory cortex (Gr(SI)), as well as the rostral portion of the taste sensory insular or claustrocortex (Cl). The rat frontal cortex also contains a second, rostral motor representation of the forelimb, trunk and hindlimb, which is somatotopically organized and may be the rat's supplementary motor area. Both of these motor representations give rise to direct corticospinal projections[21,42,51,57], some of which may make monosynaptic connections with cervical enlargement motorneurons[16]. Medial to the primary motor cortex, in cytoarchitectonic field AgM, is what appears to be part of the rat's frontal eye fields, a region which also includes the vibrissae motor representation. The somatic motor cortical output organization pattern in the rat is remarkably similar to that seen in the primate, whose primary, supplementary and frontal eye field cortical motor regions have been extensively studied.

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APPENDIX: EVOKED MOVEMENT ABBREVIATIONS

Ae	ankle extension
Af	ankle plantar flexion
De	digit extension
Df	digit flexion
Ee	elbow extension
Ef	elbow flexion
El	eyelid open
Eo	eyelid open
Hf	hip flexion
J	jaw movement
Jo	jaw open
Kf	knee flexion
L	leg movement
Ll	lower lip up
Llip	lower lip up
N	neck muscle contraction
Nk	neck muscle contraction
NR	no response
Ns	nose twitch
P	pinna movement
Pr	pronation
Pro	pronation
Pron	pronation
Rh	rhinarium (nose) twitch
Sab	shoulder abduction
Se	shoulder extension
Sf	shoulder flexion
Sr	shoulder retraction
Sup	supination
Td	tongue dorsum twitch
Tl	tail movement
To	tongue out
Tr	trunk movement
Tr	tongue right side twitch (JLT maps)
Ttip	tongue tip twitch
Tv	tongue ventral surface twitch
T5f	toe 5 flexion
V	vibrissae movement
Vc	contralateral vibrissae movement
Vi	ipsilateral vibrissae movement
We	wrist extension

REFERENCES

1 Asanuma, H., Arnold, A. and Zarzecki, P., Further study on the excitation of pyramidal tract cells by intracortical microstimulation, *Exp. Brain Res.*, 26 (1976) 443–461.
2 Asanuma, H. and Arnold, A.P., Noxious effects of excessive currents used for intracortical microstimulation, *Brain Res.*, 96 (1975) 103–107.
3 Asanuma, H. and Rosen, I., Topographical organization of cortical efferent zones projecting to distal forelimb muscles in the monkey, *Exp. Brain Res.*, 14 (1972) 243–256.
4 Benjamin, R.M. and Akert, D., Cortical and thalamic areas involved in taste discrimination in the albino rat, *J. Comp. Neurol.*, 111 (1959) 231–260.
5 Brinkman, C. and Porter, R., Supplementary motor area in the monkey: activity of neurons during performance of a learned task, *J. Neurophysiol.*, 42 (1979) 681–709.
6 Buxton, D.F. and Goodman, D.C., Motor function and the corticospinal tracts in the dog and racoon, *J. Comp. Neurol.*, 129 (1967) 341–360.
7 Castro, A.J., The effects of cortical ablations on digital usage in the rat, *Brain Res.*, 37 (1972) 173–185.
8 Castro, A.J., Tongue usage as a measure of cerebral cortical localization in the rat, *Exp. Neurol.*, 47 (1975) 343–352.
9 Chapin, J.K. and Lin, C.-S., Mapping the body representation in the SI cortex of anesthetized and awake rats, *J. Comp. Neurol.*, 229 (1984) 199–213.
10 Cheney, P.D. and Fetz, E.E., Comparable patterns of muscle facilitation evoked by individual corticomotoneuronal (CM) cells and by single intracortical microstimuli in primates: evidence for functional groups of CM cells, *J. Neurophysiol.*, 53 (1985) 786–804.
11 Clark, G., *Staining Procedures,* Williams & Wilkins, Baltimore, 1981.
12 Crowne, D.P. and Pathria, M.N., Some attentional effects of unilateral frontal lesions in the rat, *Behav. Brain Res.*, 6 (1982) 25–39.
13 Diamond, I.T., The subdivisions of neocortex: a proposal to revise the traditional view of sensory, motor, and association areas. In J.M. Sprague and A.N. Epstein (Eds.), *Progress in Psychobiology and Physiological Psychology,* Academic Press, New York, 1979, pp. 1–43.
14 Donoghue, J.P., Kerman, K.L. and Ebner, F.F., Evidence for two organizational plans in the somatic sensory-motor cortex of the rat, *J. Comp. Neurol.*, 183 (1979) 647–663.
15 Donoghue, J.P. and Wise, S.P., The motor cortex of the rat: cytoarchitecture and microstimulation mapping, *J. Comp. Neurol.*, 212 (1982) 76–88.
16 Elger, C.E., Speckmann, E.-J., Caspers, H. and Janzen, R.W.C., Cortico-spinal connections in the rat. I. Monosynaptic and polysynaptic responses of cervical motoneurons to epicortical stimulation, *Exp. Brain Res.*, 28 (1977) 385–404.
17 Gioanni, Y. and Lamarche, M., A reappraisal of rat motor cortex organization by intracortical microstimulation, *Brain Res.*, 344 (1985) 49–61.
18 Gould, H.J., Cusick, C.G., Pons, T.P. and Kaas, J.H., The relation of callosal connections to microstimulation maps of precentral motor cortex in owl monkeys, *Neurosci. Abstr.*, 9 (1983) 309.
19 Hall, R.D. and Lindholm, E.P., Organization of motor and somatosensory neocortex in the albino rat, *Brain Res.*, 66 (1974) 23–38.
20 Hardy, S.G.P. and Leichnetz, G.R., Frontal cortical projections to the periaqueductal gray in the rat: a retrograde and orthograde horseradish peroxidase study, *Neurosci. Lett.*, 23 (1981) 13–17.
21 Hicks, S.P. and D'Amato, C.J., Locating corticospinal neurons by retrograde axonal transport of HRP, *Exp. Neurol.*, 56 (1977) 410–420.
22 Hummelsheim, H. and Wiesendanger, M., Is the hindlimb representation of the rat's cortex a 'sensorimotor amalgam'?, *Brain Res.*, 346 (1985) 75–81.
23 Humphrey, D.R. and Reed, D.J., Separate cortical systems of control of joint movement and joint stiffness: reciprocal activation and coactivation of antagonist muscles. In J.E. Desmedt (Ed.), *Advances in Neurology, Vol. 39. Motor Control Mechanisms in Health and Disease,* Raven Press, New York, 1983, pp. 347–372.
24 Jankowska, E., Padel, Y. and Tanaka, R., The mode of activation of pyramidal tract cells by intracortical stimuli, *J. Physiol. (London),* 249 (1975) 617–636.
25 Jankowska, E., Padel, Y. and Tanaka, R., Projections of pyramidal tract cells to alpha-motoneurons innervating hindlimb muscles in the monkey, *J. Physiol. (London),* 249 (1975) 637–669.
26 Kolb, B., Pittman, K., Sutherland, R.J. and Whishaw, I.Q., Dissociation of the contributions of the prefrontal cortex and dorsomedial thalamic nucleus to spatially guided behavior in the rat, *Behav. Brain Res.*, 6 (1982) 365–378.
27 Krettek, J.E. and Price, J.L., The cortical projections of the mediodorsal nucleus and adjacent thalamic nuclei in the rat, *J. Comp. Neurol.*, 171 (1977) 157–192.
28 Kuypers, H.G.J.M., Cortico-bulbar connections to the pons and lower brainstem in man. An anatomical study, *Brain Res.*, 81 (1958) 364–388.
29 Kuypers, H.G.J.M., Anatomy of descending systems. In J.M. Brookhart and V.B. Mountcastle (Eds.), *Handbook of Physiology. Section 1. The Nervous System. Vol. II. Motor Control, Part 1,* Williams and Wilkins, Baltimore, 1981, pp. 597–666.
30 Kwan, H.C., MacKay, W.A., Murphy, J.T. and Wong, Y.C., Spatial organization of precentral cortex in awake primates. II. Motor outputs, *J. Neurophysiol.*, 41 (1978) 1120–1131.
31 Leonard, C.M., The prefrontal cortex of the rat. I. Cortical projection of the mediodorsal nucleus. II. Efferent connections, *Brain Res.*, 12 (1969) 321–343.
32 Macpherson, J.M., Marangoz, C., Miles, T.S. and Wiesendanger, M., Microstimulation of the supplementary motor area (SMA) in the awake monkey, *Exp. Brain Res.*, 45 (1982) 410–416.
33 Marcus, S., Zarzecki, P. and Asanuma, H., An estimate of effective spread of stimulating current, *Exp. Brain Res.*, 34 (1979) 68–71.
34 McHaffie, J.G. and Stein, B.E., Eye movements evoked by electrical stimulation in the superior colliculus of rats and hamsters, *Brain Res.*, 247 (1982) 243–253.
35 Miller, M.W. and Vogt, B.A., Direct connections of rat visual cortex with sensory, motor and association cortices, *J. Comp. Neurol.*, 226 (1984) 184–202.
36 Mori, A., Babb, R.S., Waters, R.S. and Asanuma, H., Motor effects produced by stimulation of secondary somatosensory (SII) cortex in the monkey, *Exp. Brain Res.*, 58 (1985) 440–442.
37 Murray, E.A. and Coulter, J.D., Organization of corticospinal neurons in the monkey, *J. Comp. Neurol.*, 195 (1981) 339–365.

38 Neafsey, E.J., Precentral zones related to flexion and extension in two hindlimb movements in the monkey, *Brain Res.*, 198 (1980) 453–459.
39 Neafsey, E.J., A simple method for glass insulating tungsten microelectrodes, *Brain Res. Bull.*, 6 (1981) 95–96.
40 Neafsey, E.J., Bold, E.L., Sievert, C.F., Terreberry, R.T. and Quirk, G.J., The primary motor cortex of the owl monkey: a microstimulation mapping study of the leg and arm areas, *Neurosci. Abstr.*, 9 (1983) 492.
41 Neafsey, E.J., Hurley-Gius, K. and Dimitrios, A., Efferent projections from the medial frontal cortex to the nucleus tractus solitarius, superior colliculus and olfactory bulb, *Neurosci. Abstr.*, 11 (1985) 1275.
42 Neafsey, E.J. and Sievert, C., A second forelimb motor area exists in rat frontal cortex, *Brain Res.*, 232 (1982) 151–156.
43 Norgren, R. and Wolf, G., Projections of thalamic gustatory and lingual areas in the rat, *Brain Res.*, 92 (1975) 123–129.
44 Paxinos, G. and Watson, C., *The Rat Brain in Stereotaxic Coordinates*, Academic Press, Sydney, 1982.
45 Phillips, C.G. and Porter, R., *Corticospinal Neurones. Their Role in Movement*, Academic Press, New York, 1977.
46 Price, A.W. and Fowler, S.C., Deficits in contralateral and ipsilateral forepaw motor control following unilateral motor cortical ablations in rats, *Brain Res.*, 205 (1981) 81–90.
47 Ranck, J.B., Which elements are excited in electrical stimulation of mammalian central nervous system: a review, *Brain Res.*, 98 (1975) 417–440.
48 Rizzolatti, G., Scandolara, C., Matelli, M. and Gentilucii, M., Afferent properties of periarcuate neurons in macaque monkeys. I. Somatosensory responses, *Behav. Brain Res.*, 2 (1981) 125–146.
49 Sanderson, K.J., Welker, W. and Shambes, G.M., Reevaluation of motor cortex and of sensorimotor overlap in cerebral cortex of albino rats, *Brain Res.*, 292 (1984) 251–260.
50 Sapienza, S., Talbi, B., Jaquemin, J. and Albe-Fessard, D., Relationship between input and output of cells in motor and somatosensory cortices of the chronic awake rat, *Exp. Brain Res.*, 43 (1981) 47–56.
51 Sievert, C.F., *An Anatomical, Electrophysiological, and Behavioral Comparison of the Primary and Supplementary Motor Areas of the Rat*, Ph.D. dissertation, Loyola University, Chicago, IL, 1985.
52 Sinnamon, H.M. and Galer, B.S., Head movements elicited by electrical stimulation of the anteromedial cortex of the rat, *Physiol. Behav.*, 33 (1984) 185–190.
53 Strick, P.L. and Preston, J.B., Two representations of the hand in area 4 of a primate. I. Motor output organization, *J. Neurophysiol.*, 48 (1982) 139–149.
54 Toyoshima, K. and Sakai, H., Exact cortical extent of the origin of the corticospinal tract (CST) and the quantitative contribution to the CST in different cytoarchitectonic areas. A study with horseradish peroxidase in the monkey, *J. Hirnforsch.*, 23 (1982) 257–269.
55 Welker, C., Receptive fields of barrels in the somatosensory neocortex of the rat, *J. Comp. Neurol.*, 166 (1976) 173–189.
56 Welker, W.I., Benjamin, R.M., Miles, R.C. and Woolsey, C.N., Motor effects of stimulation of cerebral cortex of squirrel monkey (*saimiri sciureus*), *J. Neurophysiol.*, 20 (1957) 347–364.
57 Wise, S.P., Murray, E.A. and Coulter, J.D., Somatotopic organization of corticospinal and corticotrigeminal neurons in the rat, *Neuroscience*, 4 (1979) 65–78.
58 Wise, S.P. and Tanji, J., Supplementary and precentral motor cortex: contrast in responsiveness to peripheral input in the hindlimb area of the unanesthetized monkey, *J. Comp. Neurol.*, 195 (1981) 433–451.
59 Woolsey, C.N., Organization of somatic sensory and motor areas of the cerebral cortex. In H.F. Harlow and C.N. Woolsey (Eds.), *Biological and Biochemical Bases of Behavior*, University of Wisconsin Press, Madison, 1958, pp. 63–82.
60 Woolsey, C.N., Settlage, P.H., Meyer, D.R., Sencer, W. Hamuy, T.P. and Travis, A.M., Patterns of localization in precentral and supplementary motor areas and their relation to the concept of a premotor area, *Patterns of Organization in the Central Nervous System. Res. Publ. Assoc. Res. Nerv. Ment. Dis.*, 30 (1952) pp. 238–264.
61 Yamamoto, T., Matsuo, R. and Kawamura, Y., Localization of cortical gustatory area in rats and its role in taste discrimination, *J. Neurophysiol.*, 44 (1980) 440–455.
62 Zilles, K., Zilles, B. and Schleicher, A., A quantitative approach to cytoarchitectonics. VI. The areal pattern of the cortex of the albino rat, *Anat. Embryol.*, 59 (1980) 335–360.